

UNIT – I

Radius of Curvature

Type – I Cartesian form (x, y)

❖ Find the radius of Curvature for the following. Curves,

1. $\sqrt{x} + \sqrt{y} = 1$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$ (Model Dec. 2017, May 2017).
2. $x^3 + y^3 = 3axy$ at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.
3. $y^2 = x^3 + 8$ at $(-2, 0)$.
4. $xy^2 = a^3 - x^3$ at $(a, 0)$ (Jan. 2016) [or] $y^2 = \frac{a^3 - x^3}{x}$ at $(a, 0)$ (May 2015).
5. $y = C \cosh\left(\frac{x}{c}\right)$ at $(0, c)$.
6. $xy = C^2$ at (c, c) .
7. $a^2y = x^3 - a^3$ at (x, y)
8. $y = 4\sin x - \sin 2x$ at $x = \frac{\pi}{2}$.
9. $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $\left(\frac{a}{4}, \frac{a}{4}\right)$ (Dec 2017, May 2017).
10. $x^2y = a(x^2 + y^2)$ at $(-2a, 2a)$.
11. $x^4 + y^4 = 2$ at $(1, 1)$.
12. $y = \frac{ax}{a+x}$ at (x, y) , Show that $\left(\frac{2P}{a}\right)^{\frac{2}{3}} = \left(\frac{y}{x}\right)^2 + \left(\frac{x}{y}\right)^2$.

Type – II Parametric form (x, y, θ) (or) (x, y, t)

❖ Find the radius of Curvature for the following. Curves,

1. $x = a(\cos\theta + \theta\sin\theta)$; $y = a(\sin\theta - \theta\cos\theta)$ is $a\theta$.

2. $x = a \cos^3 \theta; y = a \sin^3 \theta$ is $3a \sin \theta \cos \theta$. (Dec. 2017, Nov. 2016, May 2016).
3. $x = a(\theta + \sin \theta); y = a(1 - \cos \theta)$ is $4a \cos \frac{\theta}{2}$. (April/May 2018, May 2017,).
4. $x = a(\theta - \sin \theta); y = a(1 - \cos \theta)$ is $4a \sin \frac{\theta}{2}$.
5. $x = e^t \cos t; y = e^t \sin t$ is $(\sqrt{2})e^t$.
6. $x = at^2; y = 2at$.
7. $x = a[\cos t + t \sin t]; y = a[\sin t - t \cos t]$.
8. $x = 6t^2 - 3t^4; y = 8t^3$ at point t.

Centre and circle of Curvature

❖ Find the centre of Curvature of the curve,

1. $y = 3x^3 + 2x^2 - 3$ at $(0, -3)$.
2. $y = x^3 - 6x^2 + 3x + 1$ at $(1, -1)$.
3. $y^2 = 4ax$ at point (x, y) .
4. $y^3 + x^3 = 2$ at $(1, 1)$.

❖ Find the circle of Curvature of the curve,

1. $y^2 = 12x$ at $(3, 6)$.
2. $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $(\frac{a}{4}, \frac{a}{4})$.
3. $x^3 + y^3 = 3axy$ at $(\frac{3a}{2}, \frac{3a}{2})$.

Evolute: 1

❖ Find the equation of the evolute of the curve

1. a) $y^2 = 4ax$ [Parabola] b) $x^2 = 4ay$ [Parabola].
2. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ [Ellipse] (Nov. 2016, May 2015).
3. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ [Hyperbola] (May 2016).

4. $x = a(\theta - \sin\theta); y = a(1 - \cos\theta)$ [Cycloid].

5. $xy = c^2$ [Rectangular Hyperbola] (Jan. 2016).

6. $x^{2/3} + y^{2/3} = a^{2/3}$.

7. $x = a \left[\cos t + \log \tan \frac{t}{2} \right], y = a \sin t$.

8. $x = a[\cos\theta + \theta\sin\theta], y = a[\sin\theta - \theta\cos\theta]$.

Properties of Gamma functions:

1. P.T. $\sqrt{1} = 1$.

2. P.T. $\sqrt{n+1} = n\sqrt{n}$.

3. P.T. $\sqrt{1/2} = \sqrt{\pi}$.

Properties of Beta functions:

1. P.T. $\beta(m, n) = \beta(n, m)$.

2. P.T. $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta$.

3. P.T. $\beta(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy, m \& n > 0$.

4. P.T. $\beta(m, n) = \frac{\sqrt{m}\sqrt{n}}{\sqrt{m+n}}, m \& n > 0$. [Relationship between β and Γ]. (April/May 2018, May 2015).

5. P.T. $\int_0^{\pi/2} \sin^p\theta \cos^q\theta d\theta = \frac{1}{2} \left[\frac{\sqrt{\frac{p+1}{2}} \sqrt{\frac{q+1}{2}}}{\sqrt{\frac{p+q+2}{2}}} \right]$.

6. P.T. $\beta(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy$
 $= \int_0^1 \frac{x^{n-1} + x^{m-1}}{(1+x)^{m+n}} dx$

7. P.T. $\frac{\beta(m+1, n)}{m} = \frac{\beta(m, n+1)}{n} = \frac{\beta(m, n)}{m+n}$.

8. P.T. $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$.

Properties Based Problems

Type – I Integral type

1. P.T. $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.

2. P.T. $\int_0^{\frac{\pi}{2}} \sqrt{\tan \sigma} d\sigma = \frac{\sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}}}{2}$.

3. P.T. $\int_0^{\frac{\pi}{2}} \sqrt{\sin \sigma} d\sigma \times \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin \sigma}} d\sigma = \pi$ (Dec. 2017).

4. Evaluate the following:

a) $\int_0^{\frac{\pi}{2}} \sin^6 \theta d\theta$.

b) $\int_0^{\frac{\pi}{2}} \cos^5 \theta d\theta$.

c) $\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta$

d) $\int_0^{\frac{\pi}{2}} \cos^8 \theta d\theta$

e) $\int_0^{\frac{\pi}{2}} \sin^6 \theta \cos^4 \theta d\theta$

f) $\int_0^{\frac{\pi}{2}} \sin^5 \theta \cos^3 \theta d\theta$

g) $\int_0^\infty e^{-x^2} x^4 dx$

h) $\int_0^1 x^7 (1-x)^8 dx$

i) $\int_0^\infty e^{-x} x^4 dx$

j) $\int_0^\infty x^6 e^{-3x} dx$

k) $\int_0^a x^4 \sqrt{a^2 - x^2} dx$

l) $\int_0^1 x^m \left[\log \left(\frac{1}{x} \right) \right]^n dx$

m) $\int_0^{\frac{\pi}{2}} \sin^{7/2} \theta \cos^{5/2} \theta d\theta$

n) $\int_0^1 x^m (\log x)^n dx$

o) $\int_0^\infty \frac{dx}{x^4 + 1}$

p) $\int_0^1 \frac{dx}{n\sqrt{1-x^n}}$

5. Express integral $\int_0^1 x^m (1-x^n)^p dx$ in terms of Beta function and evaluate $\int_0^1 x^5 (1-x^3)^{10} dx$.

6. Show that $\int_0^1 \frac{x^n}{\sqrt{1-x^2}} dx = \frac{1.3.5 \dots (n-1)}{2.4.6 \dots n} \cdot \frac{\pi}{2}$, if n is an even integer.